

Witness-Guided Repair and Dual-Gate Checks for LLM-Generated Guard Specifications

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Abstract—Large language models synthesise code that often looks correct under everyday unit tests yet still mishandles thin specification regions, especially catch-all residuals and ordered overlaps. Consider a tier classifier whose OTHERWISE branch must return Review: an LLM candidate can pass every Reject/Pass sample while silently promoting the mid-band to Pass. A release dashboard that tracks mean pass rate alone will ship that defect.

We study headroom-conditioned catch-all repair under a SpecIR first-match oracle, together with a conditional dual-gate release check ($\text{Accept} \Leftrightarrow \text{Conf}=1 \wedge \text{Screen}=\text{pass}$). HSP-Agile implements the check on SOFL/FSF with Priority-Encoded Witness Generation, Scenario-Indexed Feedback Rendering, and a configurable static screen. FAR (False Accept Rate) is the fraction of faulty mutants that a mode incorrectly releases; PDR is the complementary rejection rate.

When test-only Conf. is already 0 ($n=13$), Semantic IR rescues all tasks (+90.7 pp); on the headroom set ($\text{Conf}.<1$, $n=114$) the paired mean is +8.2 pp (CI excludes 0). Among 18 decisive tasks, Semantic IR wins 14/18 (77.8%). The all-task +7.7 pp figure is only a supporting aggregate. On industrial-style mutants ($n=132$), Screen cuts FAR from 56.8% to 32.6%—the primary prevention claim; over the full mutant pool ($n=852$) that lift appears as FAR 8.8% to 5.0%, but HardSynthetic mutants are already saturated and do not exercise Screen. Under a frozen external hard-seed protocol (test-only Conf.<1), wrong_relop transfers at +20.6 pp ($n=79$; CI [12.7, 28.6]); HKCA09 alone contributes +18.8 pp ($n=17$; CI excludes 0). On GitHub harvest reference mutants, Screen cuts FAR from 60.4% to 11.5% (94 B2-accept/M-reject cases); published-industrial screening saturates (0 Screen-hits)—a boundary, not a null of industrial-132. The controlled SIFR estimate remains E6 on BENCH-120; external rows are protocol-transfer diagnostics, not a second headline. Test-only B2 remains the default. Escalate when Accept is required or a contractual FAR budget exists *and* repair headroom remains; Screen helps on industrial-pattern mutants but does not attain $\text{FAR} \leq 5\%$ on that slice.

Index Terms—acceptance checking, mutant rejection, specification-guided repair, ordered guards, LLM synthesis, SOFL, SMT, false accept rate

I. INTRODUCTION

A release dashboard can look healthy while a residual specification region is already broken. Large language models synthesise code that passes busy unit tests yet still mishandles catch-all OTHERWISE bands and ordered overlaps that everyday suites never sample [1]–[3]. SOFL/FSF makes those regions explicit as first-match scenarios [4]–[6]. We ask whether a team can *refuse* to ship under a specification-backed acceptance-safety bar, without full deductive verification.

Acceptance criterion. Release is safe only when every specification-derived witness matches *and* a structural screen

clears: $\text{Accept} \Leftrightarrow \text{Conf}=1 \wedge \text{Screen}=\text{pass}$. We refer to this directly as a **dual-gate acceptance check**. It is a bounded operational criterion evaluated over generated witnesses and a configured screen, not a proof of program safety. HSP-Agile realises it on SOFL/FSF with SpecIR witnesses, structured scenario-aware feedback when repair headroom remains, and a configurable static screen. FAR is the fraction of faulty mutants that a mode incorrectly releases; PDR is the complementary rejection rate.

Primary vignette (catch-all). A tier classifier rejects negative scores and passes scores at or above a floor; everything else falls into OTHERWISE and must return Review. An LLM candidate collapses that residual to Pass. Busy Reject/Pass tests succeed; the mid-band never reaches a human. Typed IR names the residual scenario (others), the same regime as all 14 E6 Conf. wins over test-only feedback (0/14 ordering repairs). The introduction uses a three-scenario skeleton with residual S_3 ; BENCH-120 hard tasks scale the same first-match shape to eight scenarios, so coded wins refer to residual index S_8 .

Secondary vignette (ordering). Ordered tax/sensor overlaps motivate first-match witnesses Φ_i and prevention mutants that reorder or drop preemption; they are not the E6 win set. A public API ACL reconstruction (`enforce_api_acl`) fail-opens on OTHERWISE while guest/admin tests pass, so Accept stays red.

Contributions. The contribution is a SpecIR interface plus a headroom-conditioned evaluation protocol for ordered-guard acceptance checking—not a new SMT decision procedure and not a proof system.

- **Priority-Encoded Witness Generation (PEWG):** a first-match SpecIR witness interface for residual and precedence regions, with the catch-all and ACL examples illustrating the Φ_i formula, with disclosed measurement corrections for fair evaluation.
- **Scenario-Indexed Feedback Rendering (SIFR):** a structured repair record that carries scenario identity, witness inputs, and expected/observed outputs into repair under headroom (collapsed catch-all residuals), not as a universal APR lift.
- **Dual-Gate Acceptance (DGA):** a release predicate and prevention *evaluation protocol* that separate witness con-

formance from a conditional structural screen under PDR/FAR (industrial-pattern mutants).

For deployment we keep test-only B2 as the default and escalate to the full loop only when Accept is required or a contractual FAR budget exists *and* repair headroom remains; the pre-hoc escalation tree is a checklist policy, not a validated predictor. Screen is a conditional second gate: it cuts FAR on industrial-style mutants (56.8% to 32.6%) but does not attain $\text{FAR} \leq 5\%$ on that slice. We evaluate mechanism (E6), prevention (E2), and saturation boundaries under stronger models or public corpora as separate evidence tracks. The controlled causal estimate for SIFR is E6 on BENCH-120; Gemini hard-seed runs, headroom-filtered external hard seeds (run_ext_hardseed_e6_v1), and public-corpus ranking runs bound saturation and transfer, not a second +7.7 pp headline. As a formal-engineering method, the dual-gate check is a bounded operational criterion under a faithful SpecIR encoding, not a theorem of program safety.

II. RELATED WORK

Closest LLM+verifier systems differ on whether witnesses respect first-match order, whether repair carries scenario structure, and whether prevention (PDR/FAR) is measured (Table I).

Proof-oriented oracles. DafnyBench and Clover target proof discharge [7], [8]; success is a theorem, not first-match region coverage. Our Conf. metric is witness agreement under SpecIR, not a proof certificate. CEGIS-style SMT loops, boundary-value suites, and concolic testing can emit satisfying inputs, but they do not by themselves encode first-match preemption ($g_i \wedge \bigwedge_{j<i} \neg g_j$) together with an *others* residual as a typed repair contract for LLM synthesis. PEWG is that SpecIR witness interface, not a new SMT decision procedure.

Verbal and agentic repair. Self-Refine, Self-Debug, and Reflexion raise generation success via critique or verbal memory [9]–[11]. That channel is LLM-as-judge style: useful, but not bound to an executable first-match oracle, so the judge can omit the catch-all scenario index. CodeT and coding agents optimise repository tasks without an FSF oracle [12]–[16]. LLM-based APR patches failing tests [17]–[21]; we evaluate unstructured test feedback as B2 and isolate typed IR in E6.

Adjacent notations and SOFL. Alloy and TLA+ encode structured constraints [22], [23]; SpecIR adapters are the intended bridge when guards are first-match and SMT-encodable. Contracts and proof-oriented LLM loops occupy complementary points [24], [25]. SOFL has a long industrial and teaching lineage [4]–[6]; railway and crossing cases motivate ordered-guard contracts but do not release public .asf1 dumps [26], [27]. Related SOFL fault-prevention work studies specification-to-program defects [28]; we target LLM repair under an executable SpecIR oracle. HSP-Agile combines first-match SMT witnesses, scenario-aware repair feedback, and a dual-gate release check evaluated with PDR/FAR. The novelty claim is therefore the ordered-guard SpecIR interface plus

TABLE I
POSITIONING AGAINST PRIOR APPROACHES (TICK = DESIGN GOAL).

Approach	Ordered Φ_i	Typed IR	Prev.	Gate
Clover / DafnyBench	—	partial	—	proof
Self-Debug / Reflexion	—	—	—	—
ChatRepair / APR	—	—	—	tests
B2 test-feedback	—	—	—	tests
B6 VerifierLoop-FSF	✓	—	—	cex
HSP-Agile / M	✓	✓	✓	Accept

headroom-conditioned typed feedback and prevention metrics, not a new decision procedure for SMT or a replacement for proof-oriented oracles.

III. APPROACH

A. A1: PEWG for First-Match Witnesses

For task t with ordered scenarios $\langle s_1, \dots, s_n \rangle$, SpecIR exposes first-match regions as a shared hinge across notations (SOFL/FSF, Mini-Z, Mini-StateMachine adapters). Each scenario $s_i = (G_i, P_i)$ pairs a guard with a postcondition; FSF selects the least i with $G_i(\mathbf{x})$ and requires $P_i(\mathbf{x}, \mathbf{y})$. A witness for s_i must satisfy s_i only after higher-priority guards have been ruled out. For scenario s_i , the witness generator solves

$$\Phi_i \triangleq \hat{g}_i(\mathbf{x}) \wedge \bigwedge_{j<i} \neg \hat{g}_j(\mathbf{x}) \quad (1)$$

with Z3 [29], [30], yielding suite $\mathcal{D}(t)$. Without the negated higher guards, a model of g_i alone may still belong to s_j for $j<i$. An *others* witness uses $\bigwedge_i \neg \hat{g}_i$ when nonempty. A prior *others*-witness bug (*others* encoded without $\neg$ prior guards) inflated false failures; we disclose it as a measurement correction and cite fixed-oracle runs as primary. Soundness is conditional on a faithful SMT encoding of SpecIR guards (integer/boolean fragments in our corpora). Ordinary random suites can undersample thin Φ_i regions, allowing residual or ordering bugs to survive green CI [3], [31].

B. A2: SIFR for Catch-All Repair

Return to the tier classifier: on a mid-band input the buggy candidate returns Pass where the oracle requires Review. The structured scenario-aware feedback record (implemented as a Semantic Feedback IR) contains:

```
{
  "violation_type": "others",
  "scenario_index": 2,
  "constraint_text": "OTHERWISE residual",
  "expected": "Review",
  "observed": "Pass",
  "suggested_fix": "return Review on catch-all"
}
```

FeedbackRenderer projects this JSON into the repair prompt. E6 isolates test_only / test_expected / semantic_ir surfaces under identical K . (A secondary ordering example uses inputs such as $(-3, -10)$ with violation_type:ordering; that trail motivates Φ_i and E2 mutants, not the E6 win profile.)

C. A3: DGA for Release

The record carries: `violation_type` (ordering, postcondition, others), `scenario_index`, `constraint_text`, concrete inputs, expected/observed outputs, and an optional `suggested_fix` hint. Conformance and release:

$$\text{Conf}(c, t) = \frac{1}{|\mathcal{D}(t)|} \sum_{\mathbf{x} \in \mathcal{D}(t)} \mathbf{1}[f_c(\mathbf{x}) = g_t(\mathbf{x})], \quad (2)$$

$$\text{Accept}(c, t) \iff \text{Conf}(c, t) = 1 \wedge P^H(c) = 0. \quad (3)$$

Mode B2 uses unstructured test pass/fail only, with no scenario index and no constraint text. Controlled E6 isolates feedback content under full M at matched $K=3$: variant A is test-only; B adds expected outputs; C is the full Semantic IR. Equal- K mode M_eq ($K=3$, `semantic_ir`, advisory gate) is the matched-budget Conf. ranking vs. B2. Strengthened E1 ($K=5$, `execution_trace_matched`, advisory gate) is a multi-factor bundle used as a near-ceiling ranking check. E14 further shows that length-matched execution traces can outperform `semantic_ir` on mean Conf. (85.4% vs. 75.1%); the E6 claim is relative to unstructured test feedback.

D. Pipeline and A4 Deployment Boundary

Algorithm 1 summarises one task. Synthesise from FSF text $\rightarrow$ check on $\mathcal{D}(t) \rightarrow$ if not Accept, render Feedback IR and repair, up to budget K ; on exhaustion return $\arg \max_k \text{Conf}(c_k, t)$. Prevention evaluation injects specification-confusion and implementation-screening mutants and scores PDR/FAR on the faulty set. The second gate is instantiated by 14 AST-level safety smells derived from SOFL fault-prevention patterns, including inverted priority, missing preemption, and unconditional success. We treat this catalogue as one reproducible screen, not a completeness claim; any severity-tagged static analyser can replace it without changing the witness check. We attribute conformance changes to witnesses and repair feedback. E2 observes the screen only within the bundled dual gate, while the reported ablation measures conformance rather than a screen-only prevention effect in isolation. Overlap density is an engineering hinge for witness generation: E3/E10 tertiles are non-monotone, so overlap alone is not a deployment trigger. The deployable levers are feedback typing (E6) and the Accept/prevention gate (E2).

For deployment, keep B2 when mean conformance and latency are the release bar. Escalate to M when the sprint requires Accept=1 or a FAR target and repair headroom remains.

IV. EXPERIMENTAL SETUP

Benchmarks. **BENCH-120**: synthetic hard FSF (5 inputs, 8 scenarios) retained by a Z3 overlap filter; primary corpus for E1/E6/E12. **E10**: removes the filter ($n=100$) to bound filter bias. **Industrial-pattern** ($n=31$): practitioner-style decision tables. **Desensitized published-industrial** ($n=28$): FSF reconstructed from published railway/ATM/banking SOFL cases [5], [26], [27]. **GitHub harvest** ($n=48$) and **HKCA09** $\rightarrow$ FSF ($n=35$) probe transfer to public specifications. Vendor .asf1

Algorithm 1 Witness-guided repair with dual-gate acceptance

Require: FSF task t , budget K , feedback mode m

```

1:  $\mathcal{D}(t) \leftarrow$  SpecIR witnesses via Z3
2:  $c^* \leftarrow \perp$ ;  $best \leftarrow 0$ 
3: for  $k = 1$  to  $K$  do
4:    $c \leftarrow$  synthesise / repair( $t$ , feedback $_{k-1}$ )
5:    $conf \leftarrow \text{Conf}(c, t)$  on  $\mathcal{D}(t)$ 
6:   if  $conf > best$  then  $c^* \leftarrow c$ ;  $best \leftarrow conf$ 
7:   end if
8:   if Accept( $c, t$ ) then return  $c$ 
9:   end if
10:  feedback $_k \leftarrow$  render( $m$ , failures on  $\mathcal{D}(t)$ )
11: end for
12: return  $c^*$  ▷ best-effort argmax Conf.

```

dumps are not redistributed; an import path exists for future NDA pilots (Table III).

Models, modes, and metrics. B1 one-shot; B2 test-feedback ($K=3$); M (strengthened bundle, $K=5$); M_eq (equal- K , $K=3$, `semantic_ir`, advisory). E6 variants A/B/C hold full M fixed and vary only feedback surface. Metrics: mean Conf.; Strict / Accept; PDR/FAR on mutants; paired wins/losses/ties with bootstrap CIs where claimed. The controlled E6 mechanism test uses ecnu-plus (Qwen-family). Existing replications include gemini-2.5-flash on hard repair seeds and GPT-4o, Claude, and DeepSeek on industrial-pattern tasks; these runs test transfer and saturation rather than replace the controlled E6 contrast. Repair temperature $T_{\text{repair}}=0$; witness budget up to 32 formal cases per task in strengthened runs.

Mutants (E2). The operators model recurring guard-generation errors: swapping or negating a scenario, reversing an output, dropping a condition, shifting a boundary, flipping a branch condition, reordering returns, dropping a branch, or reversing a dependency. Each operator makes one first-order change to a known-correct specification or implementation; uncompileable mutants are discarded. These faults exercise first-match semantics and postconditions, not just the 14 screen patterns. The implementation track contains $n=852$ mutants; pooling it with specification-confusion gives $n=1704$. PDR is the fraction of faulty candidates rejected; FAR is the fraction of faulty mutants incorrectly released under the release predicate.

Protocol. E6 isolates feedback content; E2 evaluates prevention; M_eq gives the equal- K ranking; strengthened E1 supplies near-ceiling context. Table II is the single protocol map for interpreting reported numbers.

V. RESULTS

A. RQ1: Scenario-aware repair feedback (E6)

Figure 1 and Table VI hold K and tasks fixed under full M ($K=3$, $n=120$). Variant C (full Semantic IR) reaches 86.9% mean Conf., **+7.7 pp** over test-only (A, 79.1%) and **+8.0 pp** over test+expected (B, 78.9%). Paired: **14/4/102**; 95% CI [2.3, 13.6] pp; Wilcoxon $p=0.018$. Most tasks tie near ceiling;

TABLE II

MASTER PROTOCOL TABLE. EACH HEADLINE NUMBER MUST TRACE TO ONE ROW; THIS PREVENTS MIXING K , FEEDBACK RENDERERS, GATES, AND CORPORA ACROSS CLAIMS. B6 IS A CONF-ONLY VERIFIERLOOP BASELINE ON BENCH-120; PREVENTION EQUALS B2/A2 BY DESIGN WHEN SCREEN IS OFF, SO THE DGA PREVENTION ROW COMPARES B2 / A2 / M.

Claim	Comparison	K	Feedback gate	/	Corpus	Primary structure	n	W/L/T	Δ pp	95% CI	Cliff sign
DGA / Decomp	B2 / A2 / M	3-5	Mode defaults; gate reported explicitly		E2 faulty mutants	PDR Test-Only Conf. < 1	120	14/4/102	+7.7	[2.3, 13.6]	0.08
						Test-Only Conf. = 0	114	14/4/96	+8.2	[2.3, 14.4]	0.09
						0 < test-only Conf. < 1	13	13/0/0	+90.7	[88.8, 93.3]	1.00
SIFR	A / B / C under full M	3	test-only / expected / Semantic IR		BENCH-120 and headroom strata	Δ Conf. Decisive ($W \cup L$)	101	14/4/96	-2.5	[-5.8, 0.2]	-0.03
						W/L/T, CI	18	14/4/0	—	[54.8, 91.0]%	—
External scope	A / B / C or B2 / M_eq	3	Same as source run		RealSpec / HKCA09 / GitHub / desensitized	Conf. and saturation boundary					
HTE	B2 default vs. M escalation	3-5	Pre-hoc rule, no attempt-history features		Pooled public + synthetic metadata	Escalation precision / recall					
Stress only	B1 / B2 / strengthened M	3 / 5	M uses trace-matched feedback + advisory gate		Fixed-oracle BENCH-120	Near-ceiling Conf. context					

TABLE III

EVALUATION CORPORA AND THEIR ROLES IN EACH EXPERIMENT FAMILY.

Corpus	n	Role
BENCH-120	120	Primary E1/E6/E12 (overlap-filtered)
E10 unfiltered	100	Filter-bias bound
Industrial-pattern	31	Practitioner-style decision tables
Desens. published	28	Reconstructed SOFL; all-mode 100%
GitHub harvest	48	Public .asf1; B2=M_eq
HKCA09→FSF	35	Public SOFL modules; B2=M_eq

the 14 wins form a rescue regime in which 13/14 have test-only Conf.=0. IR records name catch-all scenario *others* with arithmetic/output mismatches (Table VII). The four losses are mixed output/arithmetic residuals. Ordering/preemption defects still motivate Φ_i witnesses and E2 mutants. E14: *semantic_ir* 75.1% vs. *execution_trace_matched* 85.4% (5/19/96), so E6 is relative to unstructured feedback. Hard combo seeds on *gemini-2.5-flash* (pooled $n=80$): FULL vs. test-only +26.5pp (CI excl. 0). Field ablation (Table V): FULL beats NL-only dumps of the same facts (+28.0pp; CI excl. 0), while dropping any single IR field yields CIs that include 0, which points to structured feedback over unstructured text rather than dependence on any single field. Table IV is the primary reading: headroom tasks with test-only Conf.<1 ($n=114$) gain +8.2pp; the Conf.=0 rescue subset ($n=13$) is +90.7pp with 13/13 wins; the partial-failure band ($0 < \text{Conf} < 1$, $n=101$) does not favour Semantic IR (-2.5pp). In this band, failures are often multi-scenario or arithmetic partial mismatches where typed scenario hints can mis-steer repair;

TABLE IV

E6 HEADROOM STRATA FOR SEMANTIC IR VS. TEST-ONLY FEEDBACK. THE BROAD HEADROOM SET KEEPS THE ORIGINAL PAIRED EVIDENCE; THE COLLAPSED CONF.=0 SUBSET EXPLAINS THE MECHANISM BUT IS SMALL AND SHOULD NOT BE OVERCLAIMED. DECISIVE TASKS ARE THE NON-TIES ($W+L=18$): WIN RATE 14/18 WITH WILSON CI ON THAT RATE (NOT ON Δ CONF.).

TABLE V

FIELD STRUCTURE UNDER HARD COMBO SEEDS (GEMINI; $n=120$ CELLS).

FULL - X	Δ (pp)	W/L/T	Excl. 0
test_only	+33.2	47/8/65	Yes
test_expected	+32.6	47/8/65	Yes
ir_nl_only	+28.0	46/12/62	Yes
ir_no_scenario_id	-0.9	22/17/81	No
ir_no_expected	+5.0	29/17/74	No

the win profile (13/14 zero-A, all catch-all/arithmetic) shows that SIFR targets collapsed residuals, not every partial witness failure. Deployment rule: when test-only Conf. is already partial (>0 and <1), keep B2 rather than forcing Semantic IR. Among the 18 decisive (non-tie) tasks, Semantic IR wins 14/18 (77.8%; Wilson 95% CI [54.8, 91.0]), so the sparse absolute win count coexists with a high decisive-task win rate. The all-task +7.7pp mean is a supporting aggregate, not a claim of universal repair lift.

Catch-all mini case. On *HardSynthetic.HardCase004*, test-only Conf. is 0 while Semantic IR reaches 0.958 after naming residual scenario S_8 (*others*) with an arithmetic mismatch (*expected*≠*observed*). The same Conf.=0 $\rightarrow$ ≈ 0.88 –0.96 rescue pattern appears on several ranked catch-all candidates (e.g., *HardCase010*, *HardCase005*); IR fields point repair at the residual band that busy Reject/Pass tests never sample.

B. RQ2: Prevention under the dual gate (E2)

Prevention under the dual gate varies by corpus slice. Table VIII leads with mechanism: the *HardSynthetic* slice ($n=720$) saturates at PDR 100% / FAR 0% under B2, A2, and M, while the industrial slice ($n=132$) moves from FAR 56.8% (B2/A2) to 32.6% (M) (bootstrap 95% CI on the FAR drop: [-31.8, -17.4]pp; excludes 0). All 32 mutants that B2 accepts and M rejects are industrial, and in every case Conf.=1 under B2, A2, and M: the second gate alone separates them (Table IX). Typical operators are DRO (remove else/catch-all), MBO (off-by-one guard comparison), and WRO (wrong relational/output constant). Readers should treat the pooled 3.8pp FAR reduction as a Screen effect on industrial-style mutants; *HardSynthetic* mutants do not exercise the second gate, and pooled FAR 5.0% is *not* the release claim. On the pooled set ($n=852$), that Screen contribution yields M at PDR

Feedback variant comparison (E6)

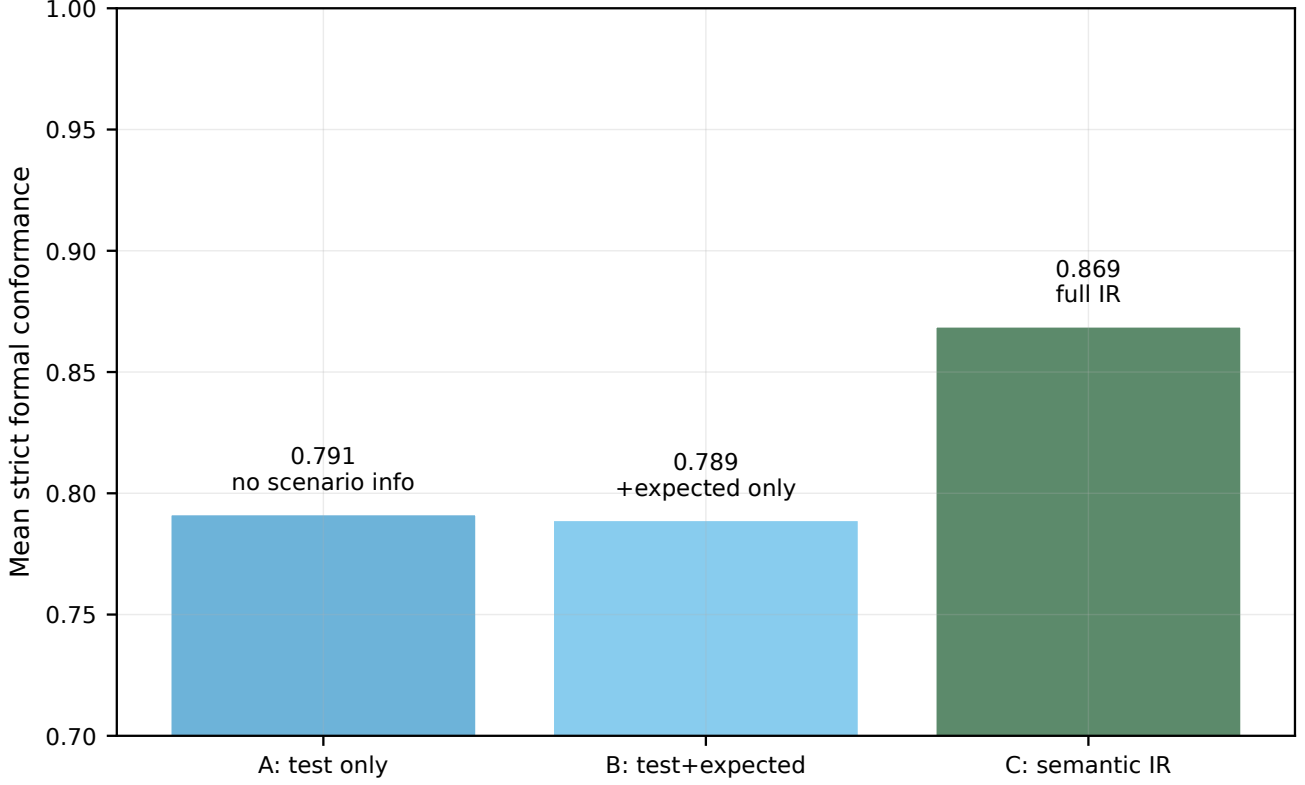


Fig. 1. E6 feedback variants under full M ($K=3$, $n=120$). Variant C (full Semantic IR) gains +7.7 pp vs. A; the 95% CI excludes 0.

TABLE VI

E6 PAIRED STATISTICS, WIN PROFILE, AND EQUAL- K CONF. RANKING.

Contrast / mode	W/L/T or n	Conf. / Δ	95% CI
E6 C-A	14/4/102	+7.7 pp	[2.3, 13.6]
E6 C-B	14/4/102	+8.0 pp	[2.2, 14.0]
Win profile	13/14 zero-A	others/arith.	—
Equal- K B2	120	97.5%	—
Equal- K M_eq	120	100.0% (+2.5)	3/0/117

TABLE VII

DEFECT CLASS AND SUPPORTING EVIDENCE MAP.

Defect class	Primary evidence	Supports
Order / preemption	Φ_i ; E2 mutants	dual-gate prevention
others/arith. rescue	E6 win coding	feedback mechanism
Saturated easy tasks	pilots ($n \geq 28$)	default B2

TABLE VIII

E2 PREVENTION DECOMPOSITION. HOW TO READ: 720/852 HARDSYNTHETIC MUTANTS SATURATE AT PDR 100% / FAR 0% UNDER EVERY FORMAL GATE, SO THEY DO NOT EXERCISE SCREEN. THE PRIMARY PREVENTION CLAIM IS THE INDUSTRIAL SLICE ($n=132$): FAR 56.8% TO 32.6% (BOOTSTRAP Δ FAR CI $[-31.8, -17.4]$ pp). POOLED FAR 8.8% TO 5.0% IS AN AGGREGATE READING ONLY, NOT THE RELEASE CLAIM. B6 IS FORMAL-ONLY (NO SCREEN): ON HARDSYNTHETIC IT MATCHES B2/A2 BY DESIGN; INDUSTRIAL B6 ROWS ARE ABSENT FROM PREVENTION_FULL_V1.

Slice	Mode	n	PDR	FAR
HardSynthetic	B2 / A2 / M	720	100.0%	0.0%
HardSynthetic	B6 (formal-only)	720	100.0%	0.0%
Industrial	B2 / A2	132	43.2%	56.8%
Industrial	M (with Screen)	132	67.4%	32.6%
Industrial	B6	—	n/a	n/a
Pooled (authoritative)	B2 / A2	852	91.2%	8.8%
Pooled (authoritative)	M	852	95.0%	5.0%

95.0% / FAR 5.0% vs. B2 91.2% / 8.8% (Fig. 2). A2 (formal witnesses, no Screen) matches B2 at 91.2% / 8.8%, so the pooled -3.8 pp FAR lift is not a larger witness budget alone. B6 and M_lite are formal-only modes like A2; by design their prevention equals B2 when Screen is off. Table VIII records B6=B2/A2 on the HardSynthetic slice (FAR 0%) and marks industrial B6 as unavailable in `prevention_full_v1`; we do not invent those FAR rows. Their role remains the Conf. / verifier-loop contrast already reported on BENCH-120.

Fixed-oracle Conf. ablations vs. strengthened M still isolate conformance (no repair -25.8 pp; no formal -17.5 pp; no pattern guard 0.0 pp on Conf.), while E2 isolates prevention. On clean reference implementations, advisory Screen matches are common (120/120), but high/critical rejection is 0/120, so the gate is low-false-kill once Conf.=1.

C. RQ3: Cross-model behavior and matched-budget evidence

The catch-all failure is not specific to one model family, but the benefit of additional feedback depends on remaining repair

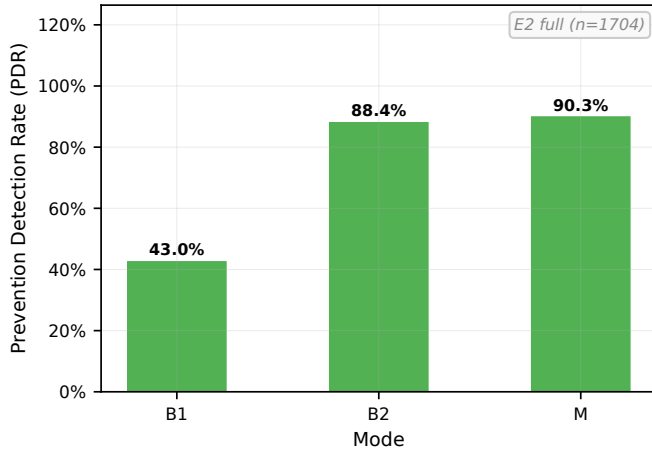


Fig. 2. E2 aggregate PDR by mode on the pooled mutant set.

TABLE IX

SCREEN-HIT VIGNETTES ON INDUSTRIAL-STYLE MUTANTS. ALL 32 CASES WHERE B2 ACCEPTS AND M REJECTS SHARE CONF.=1 UNDER B2/A2/M; THE SECOND GATE ALONE SEPARATES THEM. OPERATORS: DRO = REMOVE ELSE/CATCH-ALL BRANCH; MBO = OFF-BY-ONE COMPARISON; WRO = WRONG RELATIONAL OPERATOR.

Task	Op	Defect sketch
Ecommerce.ShipOrder	DRO	Remove else branch
Ecommerce.ShipOrder	MBO	Off-by-one in comparison
Traffic.ReportCongestion	WRO	Return wrong constant for primary output

TABLE X

CLEAN-REFERENCE SCREEN FALSE POSITIVES ON BENCH-120. THE PoC SCREEN EMITS ADVISORY MATCHES ON REFERENCE CODE, BUT NO HIGH/CRITICAL HIT WOULD REJECT A CLEAN REFERENCE IMPLEMENTATION.

Measure	Count / n	Rate
Clean reference implementations	120/120	100.0%
Any pattern match	120/120	100.0%
High or critical reject	0/120	0.0%
Critical reject	0/120	0.0%

headroom. On gemini-2.5-flash, frozen hard seeds yield +26.5 pp for scenario-aware feedback over test-only feedback (pooled $n=80$, CI excludes zero). Applying the same headroom filter to archived HKCA09/GitHub/RealSpec/industrial hard seeds (run_ext_hardseed_e6_v2) yields a transferring wrong_relop contrast: +20.6 pp on $n=79$ pairs (22/1/56; CI [12.7, 28.6]), with pooled headroom $n=186$ at +18.3 pp (CI [13.2, 23.8]). HKCA09 gemini alone: +18.8 pp ($n=17$; CI excludes 0). External impl-screening shows GitHub Screen transfer (FAR 60.4%→11.5%) beside published-industrial saturation. Unfiltered public ranking runs still often saturate; Table XI keeps those rows as ceiling diagnostics. On industrial-pattern tasks, GPT-4o and Claude both reach 100.0% with B2 while the fuller mode reaches 95.5%; DeepSeek retains more headroom (80.7% for B2 and 72.5% for M). Regression

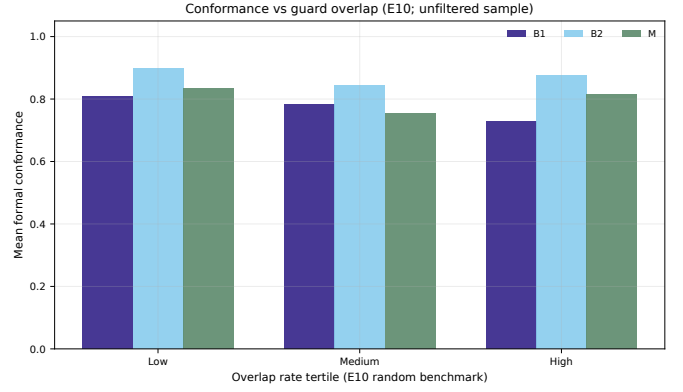


Fig. 3. E10 mean Conf. vs. overlap tertile (non-monotone).

guard: when B2 already achieves Accept=1 (or 100% mean Conf. on the release suite), do not escalate to M; the fuller loop can regress under strong models. Stronger models often saturate under ordinary test feedback. That pattern motivates conditional deployment rather than always running the full pipeline. Gate outcome Equal-K Conf. ranking (Table VI): M_eq 100.0% vs. B2 B2/A2 accept, M reject 97.5% (+2.5 pp; 3/0/117), a matched-budget check rather than a substitute for E6. Unified external equal-K (Table XV): GitHub B2=M_eq 92.0% vs. B1 83.7% (0/0/48; B2/A2 accept, M reject Accept≠mean Conf.); HKCA09 both 100%; desensitized both 100%. Strengthened E1 ($K=5$ bundle, Table XIII) sits near ceiling. E12: B2 reaches 100% on every seed; M mean 99.7% across seeds.

D. RQ4: Deployment boundary and public-corpus transfer

Canonical E3/E10 tertiles are non-monotone (Fig. 3): escalate on Accept/FAR and headroom, not on overlap density alone (Table XIV). On fixed-oracle E1, mean LLM calls for B2 (~1.18) and strengthened M (~1.16) are comparable; wall-clock means are 9.6 s (B2) vs. 10.5 s (M). The pre-hoc HTE diagnostic is deliberately weak: the best inspected tree over hard-task metadata reaches precision 0.167 and recall 0.786 against a 0.117 base positive rate. We treat HTE as a checklist policy, not as a validated predictor.

1) *Public and published corpora*: Desensitized published pilot ($n=28$): B1/B2/M_eq all 100.0%. GitHub harvest ($n=48$; run_github_harvest_v2): B2/M_eq 92.0% vs. B1 83.7% (paired 0/0/48)—B1 headroom, then B2 ties M_eq, so mean Conf. must not replace Accept. HKCA09→FSF ($n=35$): B1 74.3% / B2=M_eq 100%. Headroom-filtered HKCA09 hard-seed (wrong_relop, gemini): +18.8 pp ($n=17$; CI [3.9, 37.5]). External Screen on reference mutants (Table XVI): GitHub FAR 60.4%→11.5%; HKCA09 near-null; published-industrial saturated—industrial-132 remains primary Screen evidence. Provenance: reconstructions / public harvest; vendor NDA pending (VENDOR_INTAKE_CHECKLIST).

TABLE XI

EXTERNAL AND MULTI-SEED STATUS FOR THE FEEDBACK-ISOLATION CLAIM. SIGNIFICANT SUPPORT IS SEPARATED FROM SATURATION AND MIXED/NULL SLICES. THE HEADROOM-FILTERED EXTERNAL ROW APPLIES THE FROZEN E6_EXT_HARDSEED PROTOCOL (RETAIN PAIRS WITH TEST-ONLY CONF.<1).

Slice	Run / artifact	Key result	Use
BENCH-120 E6 anchor	run_feedback_v2	semantic_ir vs. test_only: +7.7 pp; W/L/T 14/4/102; 95% CI [2.3, 13.6]	Lead C2 mechanism, not external.
E6-Multi hard combo seeds	run_ir_combo_seed_gemini_n40_v1 + extra40_v1	240 paired cells: FULL vs. test_only +26.5 pp, CI [21.0, 32.2]; vs. test_expected +26.7 pp, CI [21.0, 32.3]	Significant support vs. unstructured feedback; do not claim uniqueness over every structured ablation.
Ext. hard-seed headroom	run_ext_hardseed_e6_v2 (HKCA09 + GitHub + RealSpec + industrial archives)	Protocol filter test-only Conf.<1: wrong_relop $n=79$, +20.6 pp, W/L/T 22/1/56, CI [12.7, 28.6]; pooled headroom $n=186$, +18.3 pp, CI [13.2, 23.8]	Protocol transfer under the same Conf.<1 filter as BENCH-120 headroom; not “external validity solved.”
GitHub harvest wrong_relop (gemini)	run_github_wrong_relop_gemini_v1	After Conf.<1 filter: $n=3$, +16.7 pp, W/L/T 1/0/2; CI includes 0 (injectable relop seeds are rare on this corpus)	Protocol transfer under Conf.<1; not a second +7.7 pp claim.
HKCA09 wrong_relop (gemini)	run_hkca09_wrong_relop_gemini_v2	Headroom filter: $n=17$, +18.8 pp, W/L/T 4/0/13, CI [3.9, 37.5] (excludes 0)	Protocol transfer under Conf.<1; supporting external C2.
Real/headroom and RealSpec E6	run_real_e6_headroom_v1; run_realspec_e6_b1fail_v1	57 paired real/headroom tasks and 30 RealSpec B1-fail tasks all saturate at 100.0% for test_only, test_expected, and semantic_ir	Saturation diagnostic when Conf. is already ceiling.
HKCA09 hard seeds (unfiltered)	run_hkca09_hard_seed_e6_v1 + v2	Without the Conf.<1 filter, families mix in sign (e.g., unfiltered wrong_relop CI includes 0)	Boundary when saturated pairs are retained; prefer the filtered row.
GitHub / HKCA09 / desensitized ranking	run_github_harvest_v2; run_hkca09_b1b2m_v1; run_desens_real_sofl_v1	GitHub: B2=M_eq=92.0% (0/0/48). HKCA09: B2=M_eq=100.0%. Desensitized pilot: all modes 100.0%.	C4 external validity and saturation/tie evidence, not feedback-isolation evidence.

TABLE XII

CROSS-MODEL INDUSTRIAL-PATTERN RESULTS ($n=31$; MEAN CONF. %).

Model	B1	B2	M
GPT-4o	95.5	100.0	95.5
Claude 4.6	95.5	100.0	95.5
DeepSeek	52.7	80.7	72.5

TABLE XIII

E1 NEAR-CEILING RANKING (FIXED ORACLE; $K=5$ BUNDLE FOR M).

Mode	Mean Conf.	Strict	Mean LLM calls
B1	84.2%	84.2%	1.00
B2	98.3%	98.3%	~1.18
M ($K=5$ bundle)	100.0%	100.0%	~1.16

2) *Public ACL interception case*: Beyond the aggregates, enforce_api_acl is an ordered API-access contract reconstructed from the public RealSpec/harvest family. The contract denies guests, requires step-up MFA for high scope, permits narrow admin/reader bands, and fails closed on OTHERWISE. An LLM-shaped candidate passes the busy guest-deny and admin-permit tests but contains two residual defects: high scope without MFA is permitted, and the catch-all branch fails open. A first-match witness ($role=2, scope=2, mfa=1$) forces the residual branch; the candidate returns permit, so Conf.<1 and the dual-gate check refuses release. This reproducible case illustrates the operational target of the acceptance check, while remaining smaller than a repository-scale industrial study.

TABLE XIV

DEPLOYMENT BOUNDARY: RELEASE BAR AND REPAIR HEADROOM.

Signal	Prefer B2	Prefer M
Default	Mean Conf./latency	—
Release	Mean Conf. OK	Accept=1
Prevention	No FAR contract	Contractual FAR budget
Headroom	Saturated endpoint	Collapsed test-only Conf. (E6)
Partial Conf.	0<test-only Conf.<1	Do not force SIFR
Strong-model Accept	B2 already Accept=1	Do not escalate
E14 structured alt.	Prefer lighter channel	Do not claim SIFR-optimal
Overlap tertile	Non-monotone	Not a trigger

TABLE XV

EXTERNAL CORPORA (MEAN CONF. %).

Corpus	B1	B2	M / M_eq
Industrial-pattern ($n=31$, gpt-4o)	95.5	100.0	95.5
Desens. published ($n=28$)	100.0	100.0	100.0
GitHub harvest ($n=48$)	83.7	92.0	92.0
HKCA09 FSF ($n=35$)	74.3	100.0	100.0
Equal- K BENCH-120	—	97.5	100.0

3) *Scope of transfer*: The controlled causal estimate remains E6 on the primary endpoint; Gemini and headroom-filtered external hard seeds support protocol transfer, while GPT-4o and Claude demonstrate saturation on industrial-pattern tasks. Public and reconstructed corpora support the deployment boundary, but they do not constitute repository-scale validation.

TABLE XVI

EXTERNAL IMPL-SCREENING (REFERENCE MUTANTS; B2/A2/M). SCREEN-HITS = MUTANTS ACCEPTED BY B2 BUT REJECTED BY M. INDUSTRIAL-132 REMAINS THE PRIMARY SCREEN EVIDENCE; EXTERNAL ROWS ARE REPRODUCIBLE EXTENSIONS OR SATURATION BOUNDARIES.

Corpus	n tasks	Screen-hits	FAR B2	FAR A2	FAR M
GitHub harvest	48	94	60.4%	60.4%	1.5%
HKCA09 FSF	35	2	50.0%	50.0%	48.6%
Published-industrial pilot	28	0	50.0%	50.0%	50.0%

Failure modes and escalation limits. Four negative results bound when *not* to escalate. First, in the partial-failure band ($0 < \text{test-only Conf.} < 1$, $n=101$), Semantic IR is -2.5 pp vs. test-only: do not force SIFR when B2 is already partially succeeding. Second, on industrial-pattern tasks GPT-4o and Claude reach 100.0% with B2 while fuller M falls to 95.5%: do not escalate when B2 already Accept=1. Third, E14 shows length-matched execution traces can beat the tested Semantic IR on mean Conf. (85.4% vs. 75.1%); SIFR is a light structured channel relative to unstructured test feedback, not the uniquely best structured renderer. Fourth, the pre-hoc HTE tree reaches precision only 0.167: treat it as a checklist reminder, not a predictor. Table XIV records these limits beside the positive escalation triggers.

VI. DISCUSSION

Practice. Match feedback depth to the release bar: B2 for mean Conf. / latency; M when Accept is required or a contractual FAR budget exists *and* repair headroom remains (E6 rescue when test-only Conf. collapses). Do not read the industrial Screen lift (FAR 56.8% to 32.6%) as attainment of $\text{FAR} \leq 5\%$ on that mutant set. External pilots support keeping B2 as the default on mean Conf. In practice, run B2 in the inner loop, run M before release when Accept matters, track FAR on a frozen mutant pack, and escalate if FAR drifts or Accept failures cluster—subject to the failure-mode limits above.

Limitations. BENCH-120 is synthetic and overlap-filtered; E10 and public pilots probe, but do not bound, transfer. Finite witnesses approximate Conf.; Conf.=1 is not a completeness claim over the full input space. Accept is sound only relative to a faithful SpecIR→SMT encoding of the guards under evaluation; encoding bugs (such as the disclosed others-witness fault) change measured Conf. and must be treated as measurement corrections. The Screen catalogue is a configurable instance, not a completeness proof over defect classes. LLM non-determinism is mitigated by $T_{\text{repair}}=0$ and paired tests; the $K=5$ E1 bundle combines several mechanisms and is used only as near-ceiling context. Published reconstructions and harvests are not production dumps; vendor NDA .asf1 pilots remain future work. The screen catalogue and mutation operators are author-selected from SOFL fault patterns; E2 estimates rejection on that constructed first-order distribution, not completeness or repository-wide safety. B6 (VerifierLoop-FSF) prevention equals B2/A2 by design when Screen is off; the BENCH-120 Conf. contrast (B6 81.1% vs. M_lite

79.0%) is conformance-only—we do not report B6 FAR without `prevention_full_v1` rows. Ceiling effects on stronger models shrink E6 deltas; report headroom when transferring claims. E14 shows that length-matched execution traces can outperform the tested scenario-aware record on mean Conf.; E6 establishes improvement over unstructured test feedback rather than superiority over every structured feedback design. B6 leads every E12 seed (100%); M mean is 99.7% across seeds.

Reproducibility. Authoritative numbers and run ids ship with the artefact package (AUTHORITATIVE_NUMBERS.md, benchmark JSON, summarisation scripts). Primary claims cite those run directories rather than ad-hoc aggregates.

VII. CONCLUSION

HSP-Agile offers headroom-conditioned catch-all repair under a SpecIR first-match oracle, plus a conditional dual-gate acceptance check for ordered-guard specifications: $\text{Accept} \Leftrightarrow \text{Conf}=1 \wedge \text{Screen}=\text{pass}$. Under collapsed test-only Conf. ($n=13$), Semantic IR rescues all tasks; on the broader headroom set the paired mean is $+8.2$ pp, with a supporting all-task aggregate of $+7.7$ pp (CI excludes 0). SIFR is not claimed superior to every structured feedback channel (E14). Prevention evidence is industrial-first: Screen cuts FAR from 56.8% to 32.6% on $n=132$ mutants; the pooled 8.8% to 5.0% reading is industrial-only and is not the release claim. Deploy B2 by default; escalate for Accept or a contractual FAR budget with headroom, knowing Screen helps but does not reach $\text{FAR} \leq 5\%$ on the evaluated industrial slice. Future work includes NDA vendor .asf1 pilots, adaptive K , and SpecIR adapters beyond SOFL/FSF.

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